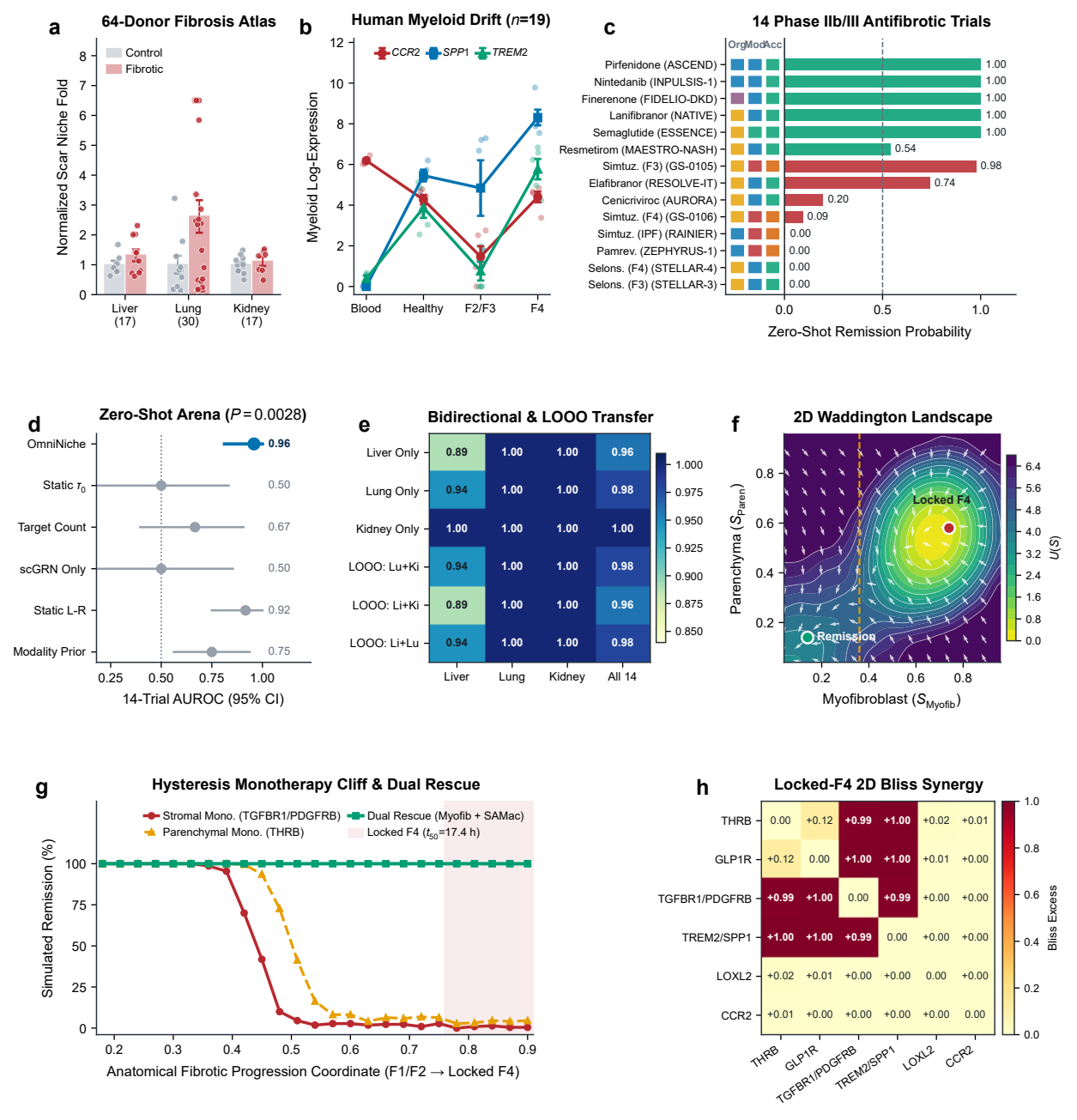

a
64-Donor Fibrosis Atlas
Control
Fibrotic
Normalized Scar Niche Fold
Liver (17)
Lung (30)
Kidney (17)
b
Human Myeloid Drift (n=19)
CCR2
SPP1
TREM2
Myeloid Log-Expression
Blood
Healthy
F2/F3
F4
c
14 Phase IIb/III Antifibrotic Trials
Org
Mod
Acc
Pirfenidone (ASCEND)
Nintedanib (INPULSIS-1)
Finerenone (FIDELIO-DKD)
Lanifibranor (NATIVE)
Semaglutide (ESSENCE)
Resmetirom (MAESTRO-NASH)
Simtuz. (F3) (GS-0105)
Elafibranor (RESOLVE-IT)
Cenicriviroc (AURORA)
Simtuz. (F4) (GS-0106)
Simtuz. (IPF) (RAINIER)
Pamrev. (ZEPHYRUS-1)
Selons. (F4) (STELLAR-4)
Selons. (F3) (STELLAR-3)
1.00
1.00
1.00
1.00
1.00
0.54
0.98
0.74
0.20
0.09
0.00
0.00
0.00
0.00
Zero-Shot Remission Probability
d
Zero-Shot Arena (P = 0.0028)
OmniNiche
Static τ0
Target Count
scGRN Only
Static L-R
Modality Prior
0.96
0.50
0.67
0.50
0.92
0.75
14-Trial AUROC (95% CI)
e
Bidirectional & LOOO Transfer
Liver Only
Lung Only
Kidney Only
LOOO: Lu+Ki
LOOO: Li+Ki
LOOO: Li+Lu
Liver
Lung
Kidney
All 14
f
2D Waddington Landscape
Locked F4
Remission
Parenchyma (SParen)
Myofibroblast (SMyofib)
U(S)
g
Hysteresis Monotherapy Cliff & Dual Rescue
Stromal Mono. (TGFBR1/PDGFRB)
Parenchymal Mono. (THRB)
Dual Rescue (Myofib + SAMac)
Locked F4 (t50=17.4 h)
Simulated Remission (%)
Anatomical Fibrotic Progression Coordinate (F1/F2 → Locked F4)
h
Locked-F4 2D Bliss Synergy
THRB
GLP1R
TGFBR1/PDGFRB
TREM2/SPP1
LOXL2
CCR2
Bliss Excess